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Perspective on thermal conductance across heterogeneously integrated interfaces for wide and ultrawide bandgap electronics

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ABSTRACT

Heterogeneous integration is important to create multi-functionality in future electronic devices. However, few thermal studies of the interfaces formed in these integrated devices have been reported before. Recently, integrated interfaces by surface-activated bonding were found to have high thermal boundary conductance, which provides a solution for heat dissipation of GaN and β-Ga_2O_3-based power electronics. Here, we review the recent progress on the interfacial thermal transport across heterogeneously integrated interfaces, including transferred van der Waals force bonded interfaces, surface-activated bonded interfaces, plasma bonded interfaces, and hydrophilic bonded interfaces. This Perspective specifically focuses on applications of thermal management strategies of electronics, especially power electronics. Finally, the challenges, such as high-throughput thermal measurements of buried interfaces, thermal property-structure relations of interfaces bonded under different conditions, theoretical understanding of interfacial thermal transport, and device demonstrations, are pointed out.

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INTRODUCTION

Accompanying the development of high-performance wide bandgap and ultrawide bandgap electronics, thermal issues, which degrade device performance and reliability, have become one of the grand challenges, which limit their potential.[1,2] As the size of electronics decreases and their power density increases, the corresponding heat flux increases, which must be addressed to meet device reliability and operational targets.[3,4] Moreover, the increased number of interfaces in these microscale and nanoscale devices impedes heat dissipation and may account for a large part of the total thermal resistance from hotspots to heat sink.[5] For instance, for power and radio frequency (RF) electronics, the interfacial thermal resistances across the device–substrate interfaces can impede effective heat dissipation.[6] Therefore, experimentally characterizing and theoretically understanding interfacial thermal transport becomes increasingly important for electronics thermal management.

Most previous studies of solid–solid interfacial thermal transport focused on interfaces fabricated by growing one material on another material.[7–12] Epitaxial growth techniques, such as molecular beam epitaxy (MBE) and metal organic chemical vapor deposition (MOCVD), can produce high-quality interfaces but are limited by lattice constant mismatch and growth procedures. Other growth techniques, such as sputter deposition, thermal evaporation, and atomic layer deposition, can grow metals and some dielectric materials but the deposited materials are usually polycrystalline or amorphous.[10] Unlike growth of interfaces, heterogeneous integration enables the fabrication of interfaces made up of two single-crystal materials regardless of their lattice constants.[13] However, thermal properties of integrated interfaces have rarely been reported even though they are important in the thermal package of wide bandgap and ultrawide bandgap electronics.[14]

Recently, surface-activated bonding (SAB) was used to directly bond single-crystal GaN and β-Ga_2O_3 with high thermal conductivity substrates, such as diamond or SiC to cool GaN and β-Ga_2O_3 based power devices.[15–18] The bonded interfaces were reported to have high thermal boundary conductance (TBC).[15–18] Additionally, plasma bonding was reported to create thermally conductive GaN–BAs interfaces for thermal management of GaN transistors.[19] This recent progress in heterogeneously integrated interfaces with high TBC will

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advance the development of both fundamental understanding of interfacial thermal transport and thermal engineering for applications, such as electronics cooling. Heterogeneously integrated interfaces provide an additional platform to study interfacial thermal transport across dissimilar interfaces, which are not achievable by direct growth. Furthermore, advances in heterogeneous integration techniques, combined with advanced thermal characterizations, can screen out the integration conditions and techniques to obtain thermally conductive interfaces and push forward the development of thermal management strategies of power and RF electronics.

This Perspective reviews recent progress in thermal studies of integrated interfaces and heterogeneous integration techniques, which are promising to fabricate thermally conductive interfaces for electronics cooling applications. We also point out the challenges and opportunities, such as high-throughput thermal characterizations of buried interfaces, thermal property-structure relations of interfaces bonded under different conditions, and device demonstrations. The focused applications are thermal management of power and RF electronics, which are based on wide and ultrawide bandgap semiconductors.

INTERFACIAL THERMAL TRANSPORT ACROSS INTEGRATED INTERFACES

Two crystals can be heterogeneously integrated by different techniques, and the resulting interfacial bonds can be van der Waals force, hydrogen bonds, ionic bonds, strong covalent bonds, or a mixture of them. The simplest heterogeneous integration method is transfer bonding. The transferred films are typically bonded with substrates by van der Waals force. For example, due to the special crystal structure [easy cleavage planes are (100) and (001)], β-Ga_2O_3 nanoscale thin crystals can be mechanically exfoliated from bulk crystals and transferred to other substrates, such as diamond.[6] Other heterogeneous integration techniques, which are capable of integrating strongly bonded and thermally conductive semiconductor interfaces, are more complicated, such as surface-activated bonding, plasma bonding, and hydrophilic bonding.

van der Waals force bonding

Typically there are no chemical bonds for van der Waals bonded interfaces. Due to the weak nature of the van der Waals bonding, the TBC values are usually very low. The experimentally measured TBC values of transferred van der Waals bonded interfaces are typically measured by time-domain thermoreflectance (TDTR) and Raman thermometry. The measured TBC values are in the range of 10–30 MW/m^2 K for different interfaces.[6,20–22] The TBC of transferred β-Ga_2O_3-diamond interface is about 17 MW/m^2 K,[6] close to those of other interfaces.[20–22] The TBC value of 17 MW/m^2 K corresponds to a Kapitza length of 129 μm in diamond. The thermal resistance from the interface equals the thermal resistance of a 129-μm-thick diamond. The low β-Ga_2O_3-diamond TBC impedes the heat dissipation from the β-Ga_2O_3 devices to the diamond substrates and prevents the devices from taking full advantage of the high thermal conductivity of single-crystal diamond.

Surface-activated bonding technique

SAB is a room-temperature bonding technique, which avoids thermal stress in high-temperature bonding due to mismatch in elastic constants and thermal expansion coefficients.[16] Figures 1(a) and 1(b) show the schematic diagram of SAB. After being polished by chemical mechanical polishing to a sub-nanometer root mean square roughness, which is typically measured by atomic force microscope (AFM) scans on an area of tens of μm^2, the surfaces of the two wafers are cleaned and activated by Ar ion beams. Then, the two wafers are bonded together by applying pressure (typically 5 MPa) and holding for 5 min. The ion activation process causes the interfacial region (a few nanometers) to be amorphous, and a small amount of Ar atoms are embedded and trapped at the interface. An optional post-annealing process can recrystallize the amorphous region and redistribute the Ar atoms.[15]

Figures 1(c)–1(f) summarize the recent progress of bonded GaN and β-Ga_2O_3 interfaces by SAB and their TBC values. The bright field images measured by a scanning transmission electron microscope (STEM) are modified from the images in the literature.[15–17,23] Figure 1(c) shows the direct bonding of single-crystal GaN with SiC wafers with a 10-min post-annealing process at 1000 °C.[15] The good crystalline interface results in a high TBC (230 MW/m^2 K), close to that of the directly grown GaN–SiC interfaces.[24] Before the annealing, an amorphous layer exists at the interfacial region and the GaN–SiC TBC is 170 MW/m^2 K. The increase in TBC after annealing could be due to the elimination of the low-thermal-conductivity amorphous layer, the increasing number of covalent bonds, or the redistribution of Ar atoms trapped at the interfacial region. Further explorations are needed to figure out the thermal property-structure relations of interfaces bonded under different conditions.

β-Ga_2O_3 can be bonded with SiC with or without an interfacial layer of Al_2O_3 by SAB.[17,18] Figure 1(d) shows the bonded β-Ga_2O_3–SiC interfaces with a 10-nm-thick Al_2O_3 interfacial layer.[17] The TBC of bonded β-Ga_2O_3–SiC interfaces with an Al_2O_3 interfacial layer is about 88 MW/m^2 K. Decreasing the thickness of the Al_2O_3 interfacial layer may potentially increase the TBC. The direct bonding of β-Ga_2O_3 and SiC without Al_2O_3 is expected to have higher TBC but the directly bonded interfaces are not uniform after annealing.[18] Further studies are needed to screen out the bonding conditions to achieve uniformly high TBC for the directly bonded β-Ga_2O_3–SiC interfaces. We emphasize that both the bonding of GaN with SiC and the bonding of β-Ga_2O_3 with SiC are wafer-scale, which facilitates scalable device fabrication.

Similarly, GaN has been demonstrated to bond with single-crystal diamond with a 4-nm-thick Si interfacial layer, as shown in Fig. 1(e).[16] The measured GaN–diamond TBC (92 MW/m^2 K) is high compared to the TBC of polycrystalline diamond grown on GaN with protective SiN_x or AlN layers by chemical vapor deposition (CVD).[16] The silicon interfacial layer may affect the electrical performance of GaN devices under high-voltage conditions. More recently, direct bonding of GaN with diamond without any interfacial layers were achieved but the TBC has not been reported.[23] The interfacial region is only 1.5 nm thick after annealing at 1000 °C. We expect a higher TBC than that of the bonded GaN–diamond interfaces with a Si interfacial layer. The lateral dimensions of these bonded GaN–diamond composites with or without silicon interfacial layers are on the order of 1–2 cm.

Plasma bonding

Another technique, which creates strong interfacial bonding, is plasma bonding. As shown in Figs. 2(a) and 2(b), a layer of Al_2O_3 is

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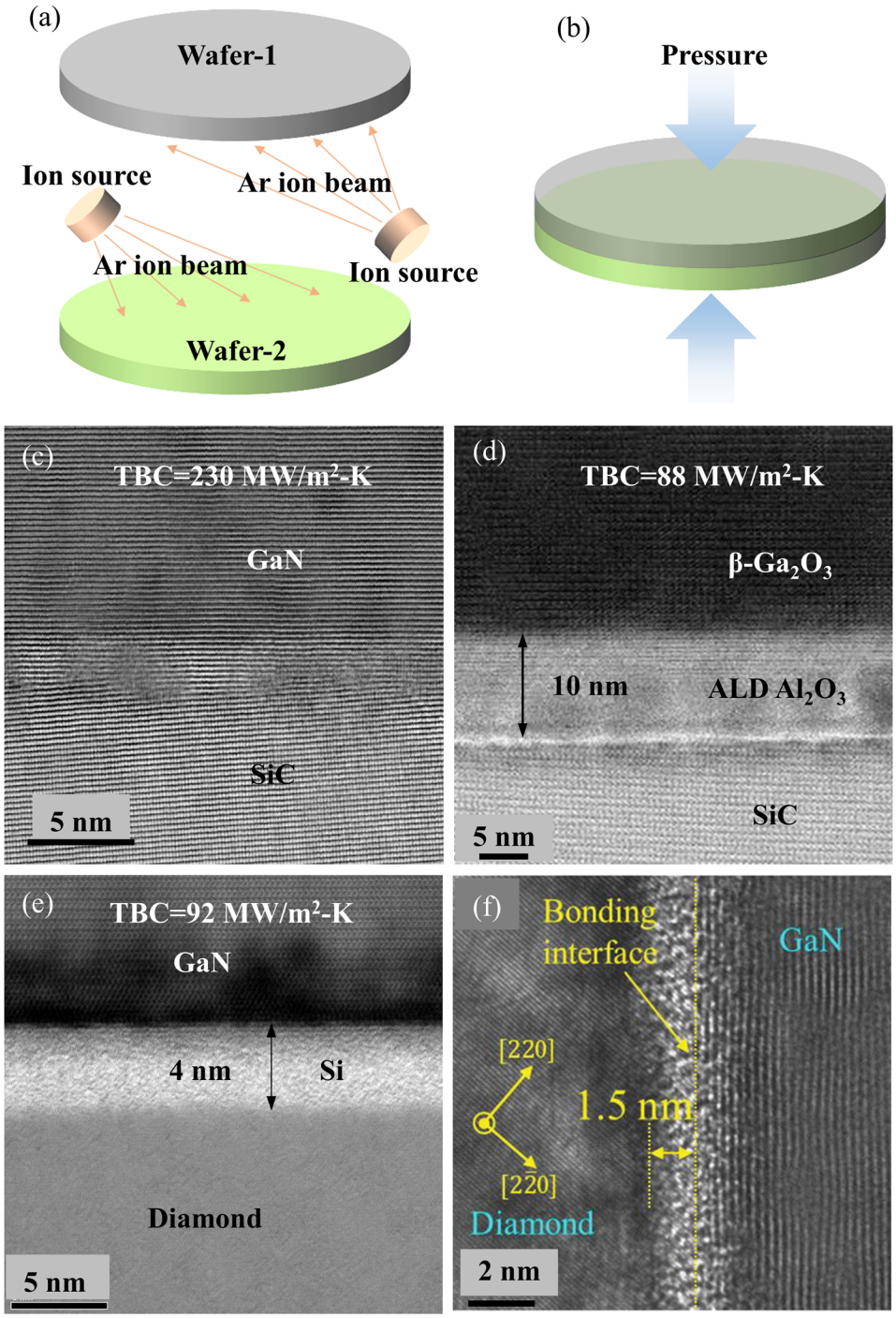


FIG. 1. Schematic diagram of surface-activated bonding technique and STEM images of bonded interfaces. (a) Ar ion beams clean and activate the wafer surfaces. (b) Two wafers are bonded together by applying a pressure (typically 5 MPa for 5 min). (c) Direct bonding of single-crystal GaN with SiC wafers with a post-annealing process at 1000 °C [Adapted with permission from Mu *et al.*, ACS Appl. Mater. Interfaces **11**, 33428–33434 (2019). Copyright 2019 American Chemical Society]. (d) Bonded β-Ga_2O_3–SiC interfaces with a 10-nm-thick Al_2O_3 interfacial layer [Adapted with permission from Cheng *et al.*, ACS Appl. Mater. Interfaces 12, 44943–44951 (2020). Copyright 2020 American Chemical Society]. (e) Bonded GaN–diamond interfaces with a 4-nm-thick Si interfacial layer [Adapted with permission from Cheng *et al.*, ACS Appl. Mater. Interfaces **12**, 8376–8384 (2020). Copyright 2020 American Chemical Society]. (f) Direct bonding of GaN with diamond with a post-annealing process at 1000 °C [Adapted with permission from Liang *et al.*, Adv. Mater. **33**, 2104564 (2021). Copyright 2021 Wiley-VCH].

deposited on the surfaces of two to-be-bonded crystals by atomic layer deposition (ALD). Then, oxygen plasma is used to activate the two surfaces before bonding the two crystals together. A post-annealing process (e.g., 500 °C in vacuum) can improve the interfacial bonding.[19] Figure 2(c) shows a STEM image of a bonded GaN–BAs interface.[19] The ALD Al_2O_3 layer is only about 2 nm thick, leading to a high TBC value (250 MW/m^2 K).[19] The thermal conductivity of high-purity BAs crystals is about half of that of diamond.[25] However, the GaN–BAs TBC is about twice as high as the TBC of GaN–diamond interfaces with additional interfacial layers reported in the literature due to the better phonon property matching between GaN and BAs and a thinner interfacial layer. The high GaN–BAs TBC facilitates heat dissipation of GaN power devices, even achieving better cooling performance at some cases than polycrystalline diamond, which is chemically

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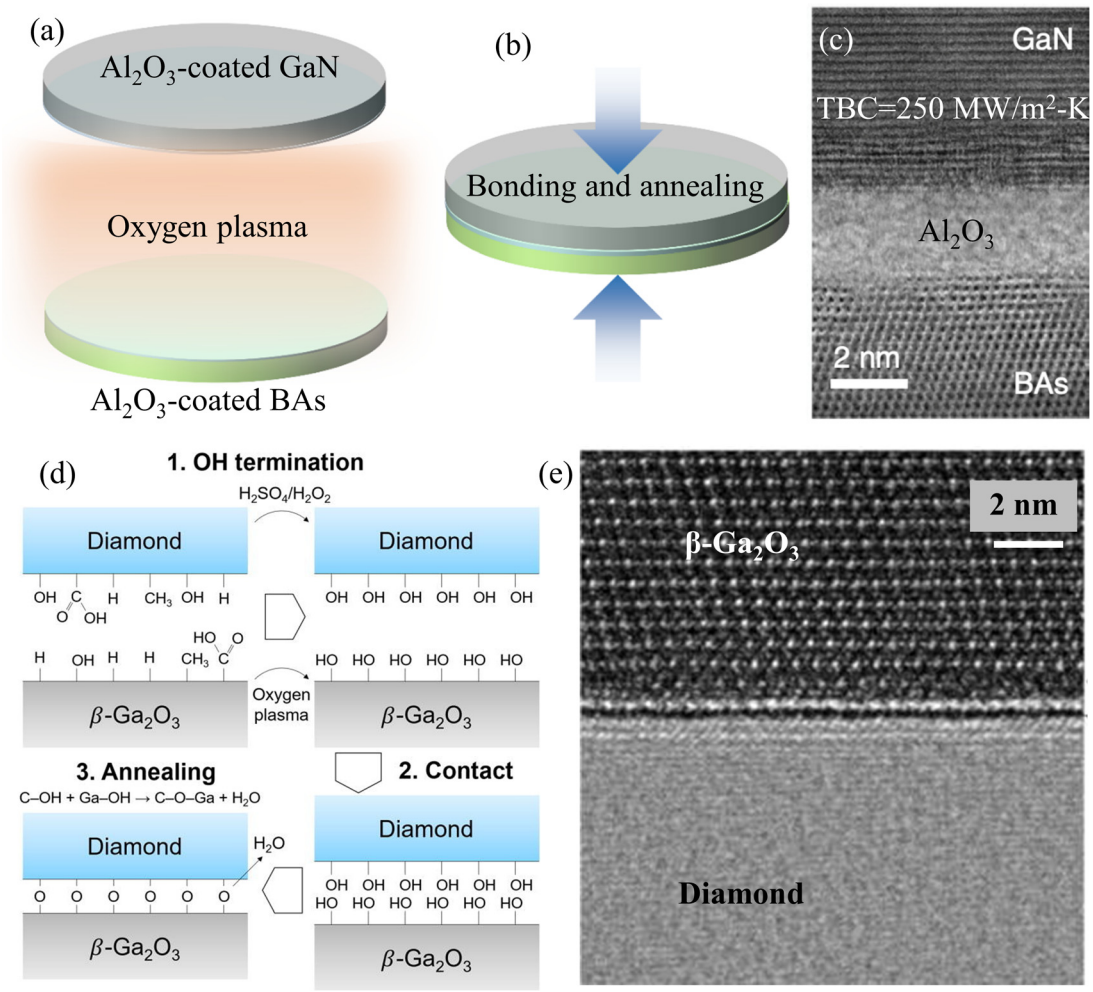


FIG. 2. Plasma bonding and hydrophilic bonding. (a) The GaN and BAs crystals are deposited with a thin layer of Al_2O_3 before applying oxygen plasma. (b) The two crystals are bonded together and annealed at 500 °C in vacuum. (c) A STEM image of the plasma-bonded GaN and BAs interface [Adapted with permission from Kang *et al.*, Nat. Electron. 4, 416–423 (2021). Copyright 2021 Springer Nature]. (d) Schematic diagram of hydrophilic bonding of β-Ga_2O_3 and single-crystal diamond [Adapted with permission from Matsumae *et al.*, Appl. Phys. Lett. **116**, 141602 (2020). Copyright 2021 AIP Publishing]. (e) A STEM image of the β-Ga_2O_3-diamond interface bonded by hydrophilic bonding technique [Adapted with permission from Matsumae *et al.*, Appl. Phys. Lett. **116**, 141602 (2020). Copyright 2021 AIP Publishing].

vapor-deposited on the back of GaN devices with a protective interfacial layer.[19] However, these BAs crystals are small (mm-scale) and they are electrically conducting. In addition, the limitations in bandgap (1.5–2 eV)[26,27] and breakdown fields will impact device limits. This inhibits the use of BAs in practical applications in operational power electronics.

For both SAB and plasma bonding techniques, interfacial layers are needed in some cases. In terms of the criteria of choosing interfacial materials, the interfacial materials need to be able to form strong mechanical bonding easily after surface activation, which guarantees that the bonding is successful. Moreover, the interfacial materials need to be easily grown as thin and uniform layers on both the materials composing the interface and have strong bonding with both materials. Atomic layer deposition is usually used to deposit thin and uniform interfacial layers before bonding. The thin interfacial layers avoid introducing large additional thermal resistances. Additionally, for applications of electronics, dielectric materials are typically preferred for the interfacial materials.

Hydrophilic bonding

Hydrophilic bonding technique is a simple and low-cost bonding technique, which can be performed without the need of a vacuum environment. Similar to other bonding techniques, the to-be-bonded surfaces need to be polished to sub-nanometer RMS roughness. As shown in Fig. 2(d), the diamond and β-Ga_2O_3 surfaces are terminated with OH by H_2SO_4/H_2O_2 cleaning and oxygen plasma bombardment.[28] Direct bonding is achieved by a thermal dehydration reaction after joining the two OH-terminated surfaces together at 250 °C for 24 h.[28] The long bonding time can be attributed to the need of water

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molecule diffusion. Further exploration is needed to understand how the water molecules diffuse and how the water molecules affect interfacial thermal transport. Figure 2(e) shows the STEM image of the β-Ga_2O_3–diamond interface bonded by hydrophilic bonding.[28] Well bonded interfaces are achieved with a sub-nanometer interfacial region and a high β-Ga_2O_3-diamond TBC is expected but not yet measured. Due to the low thermal conductivity of β-Ga_2O_3, thermal issues are grand challenges for β-Ga_2O_3-based power electronics.[29] The hydrophilic bonding technique may provide a solution to cool these devices.

Thermal modeling of power devices

To study the effects of the TBC values of these heterogeneously integrated interfaces on device thermal management, we perform calculations for the temperature profiles of devices under fully open channel conditions using the analytical solutions of steady-state heat transfer in multi-finger power devices in the literature.[30] The upside surface of the devices is assumed to be adiabatic and all the generated heat dissipates through the substrates. The bottom surface is a 50-μm SnAg die attach (33 W/m K) layer with a heat transfer coefficient of 6.5×10^5 W/m^2 K at the sink and ambient temperature of 20 °C. The dimensions of the devices are shown in Fig. 3(a). Table I summarizes the TBC values of integrated GaN-substrate and β-Ga_2O_3-substrate interfaces with different integration techniques. As shown in Fig. 3(a), a two-finger device is modeled with a size of 2×2 mm^2 and a substrate thickness of 200 μm. A typical heat source size is chosen as 4×500 μm^2 with a distance of 50 μm. The power densities are 10 W/mm for all the GaN devices and 5 W/mm for all the β-Ga_2O_3 devices. The thermal conductivity of the 1-μm GaN layer is 130 W/m K while the in-plane and cross-plane thermal conductivities of the 1-μm β-Ga_2O_3 layer are 10 and 15 W/m K.[6,20,31] The thermal conductivities of Si, SiC, BAs, and diamond substrates are 140, 320, 1000, and 2200 W/m K, respectively.[25] Here, we did not consider the effect of temperature dependence of thermal conductivity. Figure 3(b) uses a GaN-on-SiC device as an example to show the temperature profile. The integrated GaN–SiC TBC is used in the calculations.

The temperature distributions of GaN and β-Ga_2O_3 based power devices are modeled to calculate the maximum temperature rises in the devices. The generated heat in the GaN and β-Ga_2O_3 based power devices dissipates through the GaN-substrate and β-Ga_2O_3-substrate interfaces. As a result, the GaN-substrate and β-Ga_2O_3-substrate TBC affect the maximum temperature rises. The substrates are Si, SiC, BAs, and diamond. Figures 3(c) and 3(d) show the effects of GaN-substrate TBC or β-Ga_2O_3-substrate TBC on the calculated maximum temperature rises in the devices with different substrates. The red squares show the TBC values of integrated GaN–SiC interfaces, GaN–BAs interfaces, GaN–diamond interfaces, β-Ga_2O_3–SiC interfaces, and β-Ga_2O_3–diamond interfaces. With these TBC values, we can compare the corresponding maximum temperature rises of GaN and β-Ga_2O_3 power devices integrated with different substrates. Lower maximum temperature rises mean better cooling performance.

For small TBC values, the maximum temperature rises increase significantly as the TBC values decrease. For TBC values larger than 100 MW/m^2 K, the maximum temperature rises reach a plateau. For a certain TBC value, high thermal conductivity substrates can effectively extract heat from the devices and result in significantly reduced maximum temperature rises. Single-crystal diamond has the highest thermal conductivity among natural materials. A GaN-diamond TBC larger than 100 MW/m^2 K or even 150 MW/m^2 K can enable the best cooling performance of any heat spreaders according to Fig. 3(c). The direct bonding of GaN and single-crystal diamond without any Si interfacial layer is promising to achieve high TBC value.[23]

CHALLENGES AND OUTLOOKS

Heterogeneous integration is important for thermal packaging of wide bandgap and ultrawide bandgap electronics but the thermal characterizations of integrated interfaces are still limited to the interfaces mentioned above. One of the challenges is the lack of high-throughput thermal characterization techniques to measure TBC of buried interfaces, especially nonmetal-nonmetal interfaces. In the literature, only ~50 interfaces have been measured so far and most of them were metal-nonmetal interfaces and measured by TDTR.[32] Integrated interfaces are typically nonmetal-nonmetal interfaces. For TDTR measurements, a layer of metal transducer needs to be deposited on top of the samples. The multi-layer structure introduces additional unknown parameters in the data fitting. Moreover, if the interfaces are buried deeply, the TDTR sensitivities of the TBC values are small due to the limitation of the thermal penetration depth of TDTR measurements. For example, for a material with thermal conductivity of 1 W/m K, TDTR is only able to probe several hundred nanometer beneath the sample surface at room temperature. Even for high thermal conductivity materials, TDTR can only probe a few micrometers beneath the sample surface. Therefore, it is necessary to develop high-throughput thermal characterization techniques to study interfacial thermal transport across integrated interfaces. Thermal metrology without the additional transducer layers or with the capability to probe deeply into the samples facilitates fast and reliable thermal characterizations of integrated interfaces.

Heterogeneous integration allows an additional degree of freedom to fabricate interfaces made up of two single-crystal materials and fabricate devices while addressing the need for heat dissipation directly at the die level. The interfacial structure-TBC relations of integrated interfaces remain unclear since the detailed interfacial structures are determined by integration conditions such as bonding techniques, interfacial bond types, implanted ion types, and post-processing. For example, post-annealing can crystallize amorphous materials near the interfaces and redistribute Ar ions at the interface for surface-activated bonded interfaces. An open question is how surface roughness of the wafers before bonding affects the TBC of bonded interfaces. The activation process is the key for wafer bonding but how all the parameters used in activation affect interfacial thermal transport across the bonded interface remains unknown. Optimizing bonding parameters used in these integrated techniques, combined with feedback from high-throughput thermal characterizations, facilitates achieving thermally conductive interfaces.

Some conventional well-grown interfaces can be considered as abrupt and clean interfaces, which are close to the ideal interfaces assumed in conventional computational tools for TBC calculations.[10] Most grown interfaces have complicated interfacial structures such as atomic diffusion layers, transition layers, and defects. The grown materials which form the interfaces are polycrystalline in some cases, which can affect TBC due to different phonon properties with different crystal orientations. Heterogeneously integrated interfaces also have complicated interfacial structures such as amorphous layers, trapped ions, element diffusion layers, and even additional interfacial layers in some

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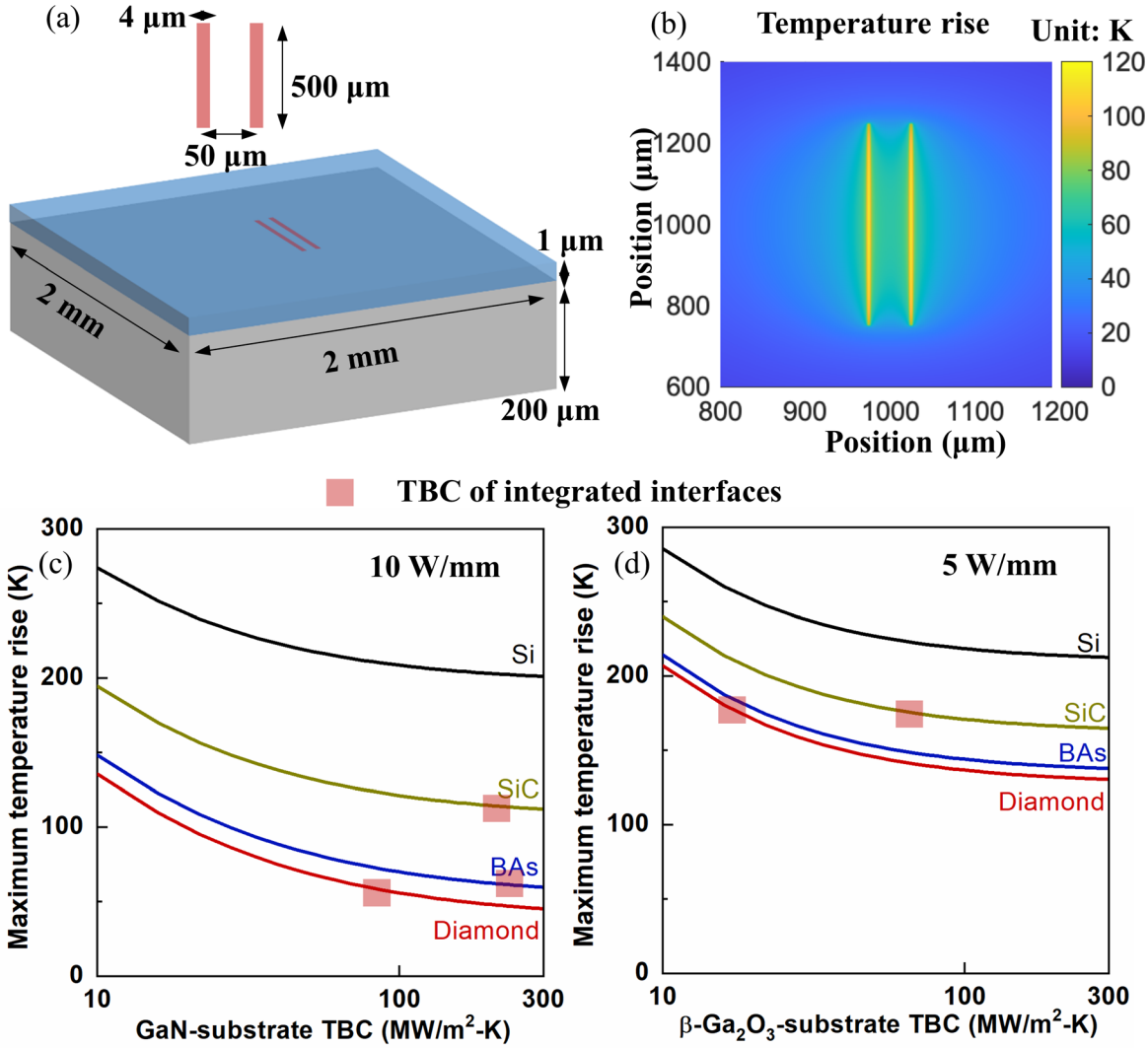


FIG. 3. Power device modeling. (a) The geometry of the modeled two-finger devices. The device size is $2 \times 2\,\text{mm}^2$ and the thickness of the substrates is 200 μm. The heat source size is $4 \times 500\ \mu\text{m}^2$ and the distance is 50 μm. (b) The temperature rise distribution of a GaN-on-SiC device with the integrated GaN–SiC TBC. (c) Effect of GaN-substrate TBC on the maximum temperature rise of power devices. The thermal conductivity of the 1-μm GaN device is 130 W/m K and the used power density is 10 W/mm for all GaN devices. The thermal conductivities of Si, SiC, BAs, and diamond substrates are 140, 320, 1000, and 2200 W/m K, respectively.[25] (d) Effect of β-Ga_2O_3-substrate TBC on the maximum temperature rise of power devices. Power density is 5 W/mm for all β-Ga_2O_3 devices and the in-plane and cross-plane thermal conductivities of the 1-μm β-Ga_2O_3 layer are 10 and 15 W/m K, respectively.[6,20,31]

TABLE I. TBC of integrated GaN-substrate and β-Ga_2O_3-substrate interfaces with different integration techniques. SAB is surface-activated bonding.

Reference	Interfaces	TBC (MW/m^2 K)	Integration techniques	Post-process
Ref. 15	GaN–SiC	170	SAB	N/A
Ref. 15	GaN–SiC	230	SAB	Annealing
Ref. 16	GaN–diamond	92	SAB	N/A
Ref. 19	GaN–BAs	250	Plasma-bonding	Annealing
Ref. 17	β-Ga_2O_3–SiC	88	SAB	N/A
Ref. 6	β-Ga_2O_3–diamond	17	Transfer bonding	N/A

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cases. The disorder at the interfacial region may impede phonon transport and lead to a reduced TBC. However, if well-designed, the additional thin interfacial layer can work as a phonon bridge to enhance TBC for some interfaces,[33] and defects at the interfacial region can redistribute phonon distributions and possibly enhance TBC.[34]

In terms of the theoretical understanding of interfacial thermal transport, first-principles calculations of TBC are still missing.[10] Existing models for interfacial thermal transport have their disadvantages. Diffuse mismatch model (DMM) and acoustic mismatch model (AMM) cannot consider the detailed interfacial structures and anharmonic contributions.[10] Atomic Green's function (AGF) includes the interfacial details but is difficult to include anharmonic contributions.[35] The development of molecular dynamics (MD) simulations can calculate TBC with considering the interfacial details and anharmonic contributions but are based on classical mechanics and cannot directly include quantum effects. Moreover, the results of MD simulations depend strongly on the used interatomic potentials.[36]

Recently, theoretical and experimental studies reported the existence of interfacial modes, suggesting the revision of conventional theories based on the transmission concept of bulk phonon modes such as DMM, AMM, and AGF.[37] A possible way to further push forward the theories of interfacial thermal transport is to take advantage of machine learning. The built TBC database of integrated interfaces under different bonding conditions, enabled by high-throughput thermal and materials characterizations, can provide data inputs for machine-learning and possibly stimulate improved theoretical understanding of interfacial thermal transport. Additionally, detailed materials characterizations of the integrated interfaces need to be performed extensively to understand the interfacial structures such as atomic arrangement, crystalline, defects, disorder, and element distributions.

One of the final goals to understand interfacial thermal transport across integrated interfaces is to develop advanced thermal management strategies for real-world power devices. Applying integrated thermally conductive interfaces to power devices remains an open area.[19] Device demonstrations and characterizations of cooling performance will be exciting but challenging. For example, for GaN-on-diamond power devices, in addition to thermal considerations, the electrical properties of GaN devices might be affected by the bonded interfacial layers and the mechanical stresses of the bonded interfaces are affected by any high temperature processes in device fabrications.[38,39] Furthermore, accurate measurements of the hotspot temperature near the gate are challenging since the hotspot are located within tens of nanometers in size and near the surface.

SUMMARY

This Perspective reviews recent progress in interfacial thermal transport across heterogeneously integrated interfaces fabricated by transfer bonding, surface-activated bonding, plasma bonding, and hydrophilic bonding. Integrated interfaces with high TBC facilitate thermal management of power and RF electronics, which are based on wide and ultrawide bandgap semiconductors. By modeling the temperature distributions of power devices with an analytical solution, we obtained the maximum temperatures of GaN and β-Ga_2O_3 based power devices with different substrates and device-substrate TBC. Moreover, we compared the cooling performance of power devices with TBC of integrated interfaces. Finally, we discussed the challenges and outlooks in thermal characterizations of buried interfaces, understanding interfacial thermal transport across integrated interfaces with different bonding conditions and techniques, and device demonstrations by applying bonding techniques.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts of interest to disclose.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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